

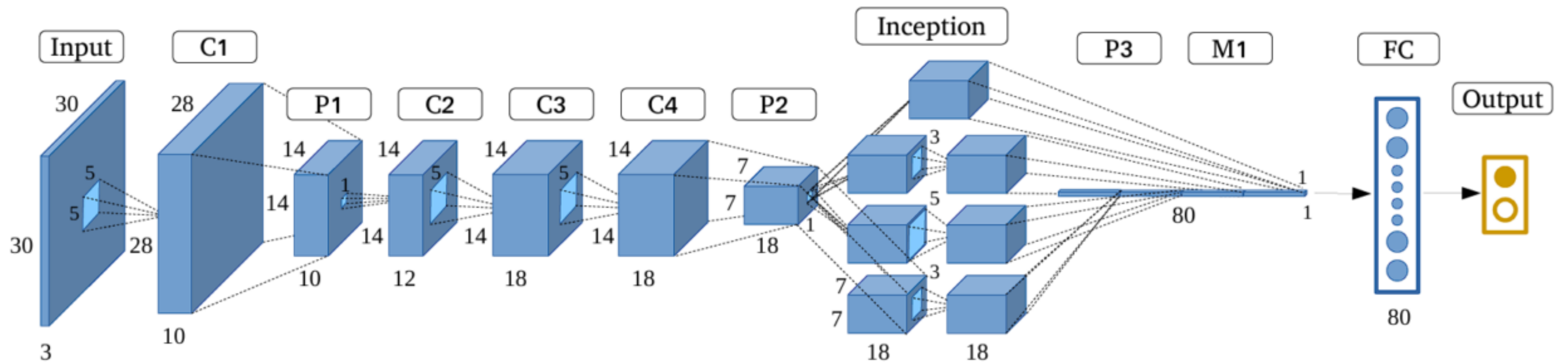
GoogleNet (Inception v1)

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GoogleNet (Inception v1)

- Developed by Google in 2014.
- Won ILSVRC 2014 (ImageNet Large Scale Visual Recognition Challenge).
- Deep convolutional NN architecture designed for efficient image classification
- It is a 22-layer CNN designed for accuracy and efficiency.
- Uses 9 inception modules to process features at multiple scale.
- Only 5 million parameters.

Architecture Overview



Architecture Overview

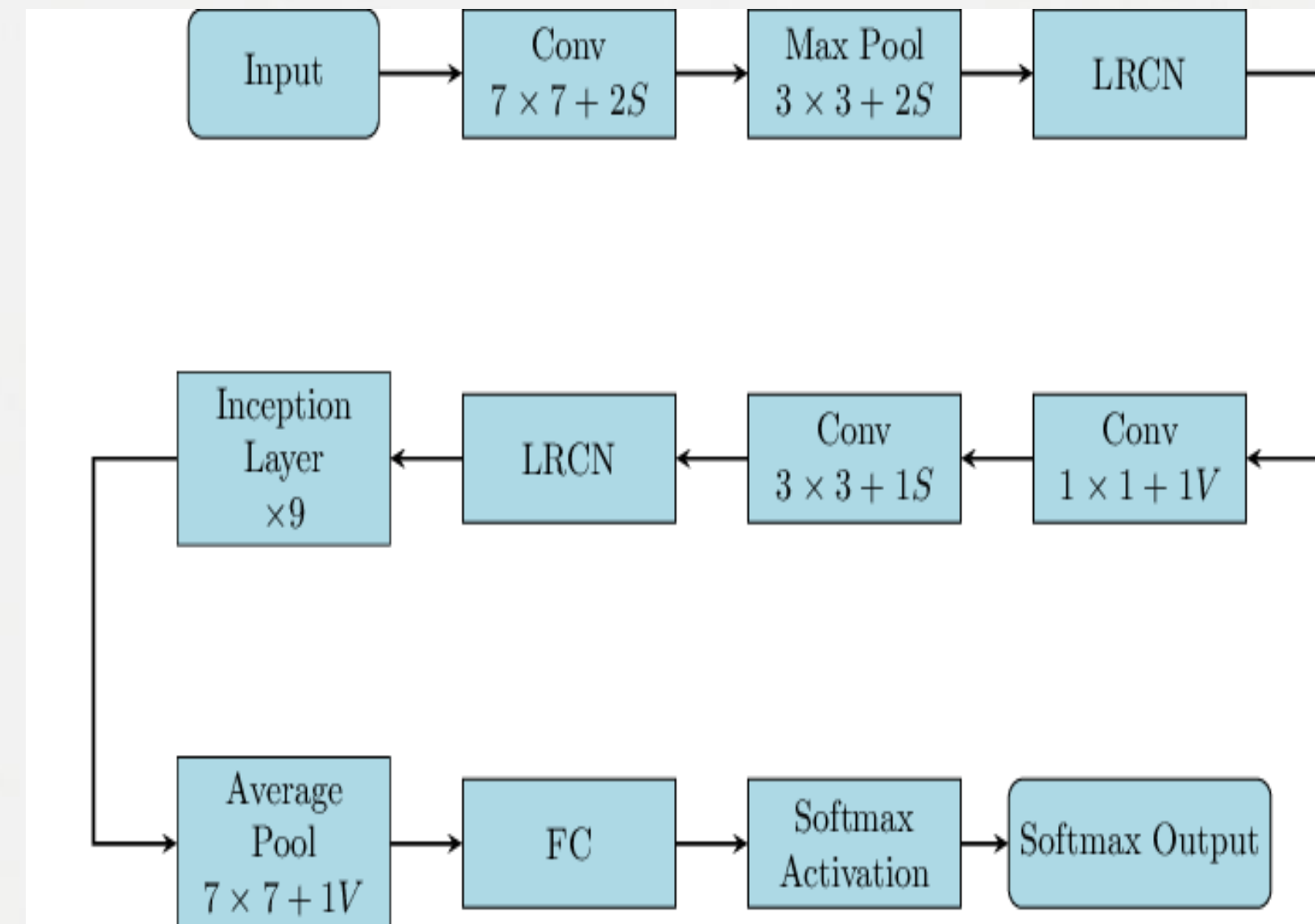
type	patch size/ stride	output size	depth	# 1×1	# 3×3 reduce	# 3×3	# 5×5 reduce	# 5×5	pool proj	params	ops
convolution	$7\times 7/2$	$112\times 112\times 64$	1							2.7K	34M
max pool	$3\times 3/2$	$56\times 56\times 64$	0								
convolution	$3\times 3/1$	$56\times 56\times 192$	2		64	192				112K	360M
max pool	$3\times 3/2$	$28\times 28\times 192$	0								
inception (3a)		$28\times 28\times 256$	2	64	96	128	16	32	32	159K	128M
inception (3b)		$28\times 28\times 480$	2	128	128	192	32	96	64	380K	304M
max pool	$3\times 3/2$	$14\times 14\times 480$	0								
inception (4a)		$14\times 14\times 512$	2	192	96	208	16	48	64	364K	73M
inception (4b)		$14\times 14\times 512$	2	160	112	224	24	64	64	437K	88M
inception (4c)		$14\times 14\times 512$	2	128	128	256	24	64	64	463K	100M
inception (4d)		$14\times 14\times 528$	2	112	144	288	32	64	64	580K	119M
inception (4e)		$14\times 14\times 832$	2	256	160	320	32	128	128	840K	170M
max pool	$3\times 3/2$	$7\times 7\times 832$	0								
inception (5a)		$7\times 7\times 832$	2	256	160	320	32	128	128	1072K	54M
inception (5b)		$7\times 7\times 1024$	2	384	192	384	48	128	128	1388K	71M
avg pool	$7\times 7/1$	$1\times 1\times 1024$	0								
dropout (40%)		$1\times 1\times 1024$	0								
linear		$1\times 1\times 1000$	1							1000K	1M
softmax		$1\times 1\times 1000$	0								

Input

- Image: 224*224*3 RGB image
- Preprocessing: normalization and data augmentation.

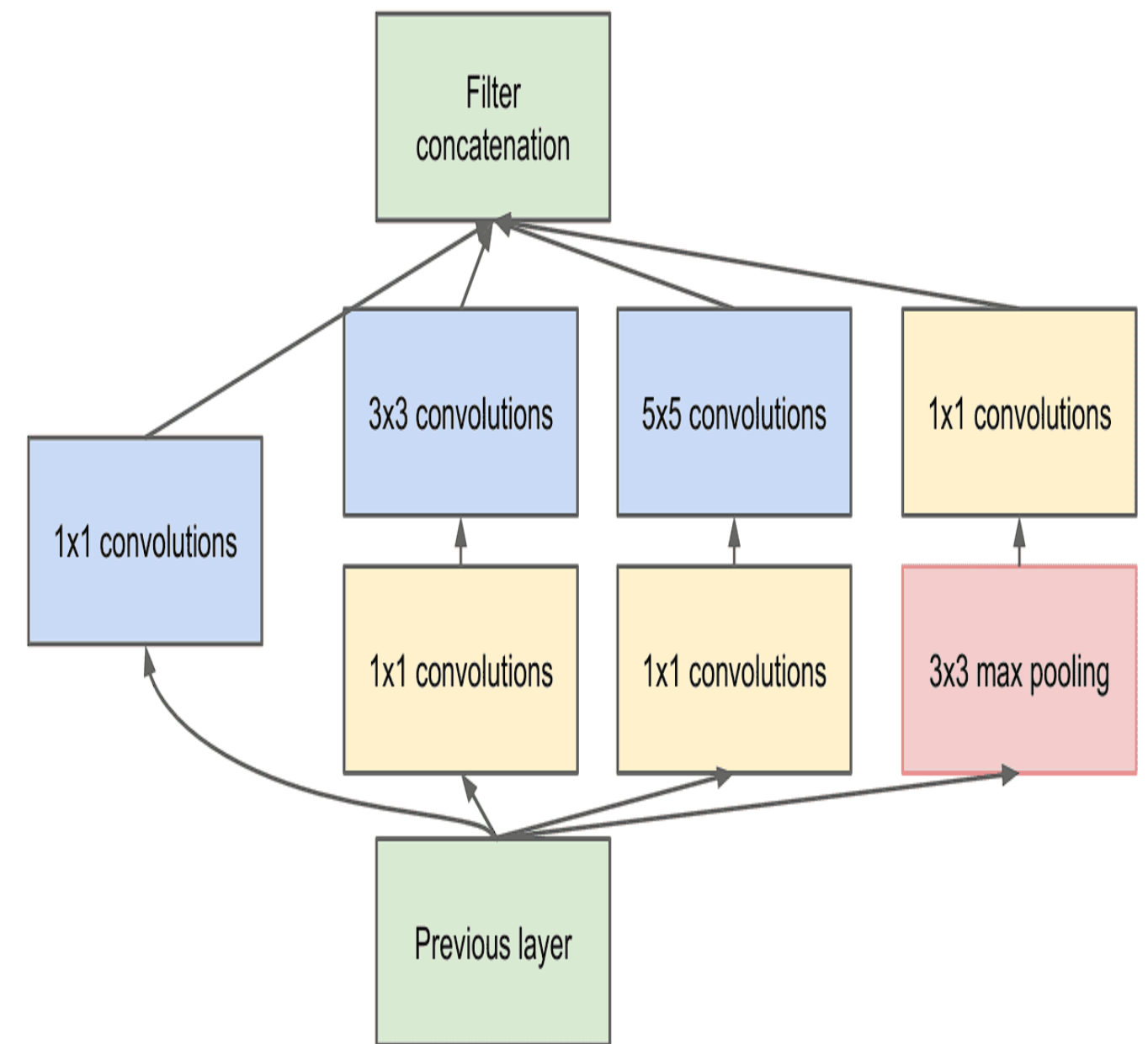
Initial Layers

- Conv1: 7*7 convolution, stride 2, extracts low level features.
- Max Pooling: 3*3 stride 2, reduces spatial size.
- Conv2: 1*1 and 3*3 convolutions, more feature extraction.
- Max Pooling: Again reduces size.



Inception Module

- Capture multi-scale features in a single layer.
- 1 x 1 convolutions reduce dimensions, reduces no. of parameters.
- Output from all 4 branches are concatenated along the depth axis.
- Input goes to all 4 branches in parallel.
- Each branch extracts features at different scales.
- Outputs are concatenated and passed to next module



Final Layers

- Inception5b output as input to final layer.
- Global average pooling converts feature maps to $1 \times 1 \times 1024$
- Dropout (0.4), reduces overfitting
- Softmax, outputs class probabilities



How it Works ?

- Image passes through initial conv layers, extracts low level features.
- Goes through Stacked Inception Modules, extracts multi-scale features.
- Intermediate auxiliary classifiers, provide extra gradients.
- Final inception module output, global average pooling → compressed feature vector.
- Dropout reduces overfitting.
- Predict class probabilities via softmax.



Advantages

- Fewer parameters.
- Multi-scale Feature extraction.
- Reduces dimensionality before heavy convolutions.

Disadvantages

- Complex architecture.
- Not as flexible to adapt for other tasks.
- Understanding what each branch learns is more complex